

Joyce Chang

CONTACT INFORMATION

Syracuse University
Department of Economics
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DOCTORAL STUDIES

Syracuse University
PhD, Economics, Expected completion May 2025

DISSERTATION COMMITTEE AND REFERENCES

Thomas Pearson (Primary Advisor) Assistant Professor of Economics Syracuse University tpears01@syr.edu	Hugo Jales (Syracuse Placement Officer) Associate Professor of Economics Syracuse University hbjales@syr.edu +1 (315) 443-3114
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Maria Zhu
Assistant Professor of Economics
Syracuse University
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PRIOR EDUCATION

The University of Texas at Austin Master of Arts in Economics	2017
National Tsing Hua University Bachelor of Science in Quantitative Finance	2014

CITIZENSHIP

United States and Taiwan

FIELDS

Primary Field: Labor Economics
Secondary Fields: Public Economics and Urban Economics

TEACHING EXPERIENCE

Public Economics (undergrad, Syracuse University) Teaching Assistant to Prof. Jeffrey Kubik	2024
Topics in Financial Economics (undergrad, Syracuse University) Teaching Assistant to Prof. Jeffrey Kubik	2020-24
Topics in Financial Economics (undergrad, Syracuse University) Teaching Assistant to Prof. Perry Singleton	2020- 21;23
Economics of Migration (undergrad, Syracuse University) Teaching Assistant to Prof. Thomas Pearson	2022
Indian Economy (undergrad, Syracuse University) Teaching Assistant to Prof. Devashish Mitra	2022
Intermediate Macroeconomics (undergrad, Syracuse University)	2019-20

	Teaching Assistant to Prof. Andrew Jonelis	
	Introductory Macroeconomics (undergrad, Syracuse University)	2019
	Teaching Assistant to Prof. Andrew Jonelis	2018-19
	Teaching Assistant to Prof. Derek Laing	
RESEARCH EXPERIENCE	Research Assistant to Prof. Thomas Pearson	2024
	Syracuse University	
	Research Assistant to Prof. Perry Singleton	2020
	Syracuse University	
	Research Assistant to Dr. Henghsiu Tsai	2013-16
	Academia Sinica	
AWARDS	Graduate Assistantship, Syracuse University	2018-24
CONFERENCE & SEMINAR PRESENTATIONS	Syracuse University Graduate Student Workshop	2024
	Syracuse University Labor Group Seminar	2022-24
	Institute of Mathematical Statistics Asia Pacific Rim Meeting	2014

RESEARCH PAPERS

“Remote Work and People with Disabilities” (Job Market Paper)

This paper investigates the heterogeneous effect of access to remote work on the labor market outcomes of people with disabilities. I use a two-way fixed effects strategy exploiting the quasi-random variation in broadband access across the United States and industry-specific shocks in the demand for remote workers. I find that remote work access increases employment, hours worked, and wages of people with disabilities. This paper sheds light on how expanding remote work opportunities could play a crucial role in improving economic outcomes for people with disabilities.

“Using Credible Intervals and Credible Regions to Detect Differential Item Functioning in IRT Models” (with Ya-Hui Su and Henghsiu Tsai)

Quantitative Psychology, pp. 297-304, 2018

Differential item functioning (DIF) occurs when individuals from different groups with the same level of ability have different probabilities of answering an item correctly. In this paper, we develop a Bayesian approach to detect DIF based on the credible intervals within the framework of item response theory models. Our method performed well for both uniform and non-uniform DIF conditions in the two-parameter logistic model. The efficacy of the proposed approach is demonstrated through simulation studies and a real data application.

“A Three-Parameter Speeded Item Response Model: Estimation and Application” (with Henghsiu Tsai, Ya-Hui Su, and Edward M. H. Lin)

Quantitative Psychology, pp. 27-38, 2016

When given time constraints, it is possible that examinees leave the harder items till later and are not able to finish answering every item in time. In this paper, this situation was modeled by incorporating a speeded-effect term into a three-parameter logistic item response model. Due to the complexity of the likelihood structure, a Bayesian estimation procedure with Markov chain Monte Carlo

method was presented. The methodology is applied to physics examination data of the Department Required Test for college entrance in Taiwan for illustration.

**RESEARCH IN
PROGRESS**

“The Differential Labor Market Impacts of Remote Work by Race and Ethnicity” (with Thomas Pearson)

This paper investigates if certain groups of minority workers are disproportionately harmed by negative remote work shocks. As Hispanic and Black workers are more concentrated in industries and occupation that are intensive in face-to-face interactions, such as the restaurant industry, they may experience larger employment losses when remote work becomes more prevalent.

**TECHNICAL
SKILLS**

Stata, R, Fortran, LaTeX, MS Office

LANGUAGES

Mandarin (native), English (fluent)

Last updated: February 2025